



SAWTOOTH WAVE GENERATOR

Using 4-Bit DAC

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A sawtooth (staircase) wave generator is a circuit that converts a digital binary sequence into a rising analogue voltage ramp. With a 4-bit R-2R

ladder digital-to-analogue converter (DAC), the output is not perfectly continuous; instead, it comprises 16 discrete voltage levels (0-15), corresponding to the 2^4 digital states. As a

and grounding issues in analogue systems.

The generated stepped ramp is useful for timing, testing, waveform generation, ADC evaluation, and basic communication applications. Economical and simple in design, the system is well-suited for educational use and can be easily implemented on a breadboard without requiring a microcontroller or specialised components.

Circuit and working

This circuit comprises three main stages: a clock source, a 4-bit binary counter, and an R-2R ladder DAC, with an optional op-amp for buffering or gain. A 555 timer IC, configured in astable mode, generates clock pulses (around 700Hz) that drive the counter. The outputs (Q0-Q3) are applied to the R-2R ladder network, which converts the digital count (0-15) into corresponding stepped analogue voltages. A $0.1\mu\text{F}$ capacitor can be added at the output for slight smoothing, and an op-amp can be included for buffering or amplification, if needed.

Fig. 1 shows the circuit diagram of the sawtooth generator built around NE555 (U2) and 74LS93 (U3). Capacitor C1 ($1\mu\text{F}$) is used for noise decoupling. The oscillation frequency is set by R1 ($1\text{k}\Omega$), R2 ($10\text{k}\Omega$), and C2 ($0.1\mu\text{F}$). The pulses from pin 3 of the NE555 are fed to the clock input (CP0) of the 74LS93, a 4-bit ripple counter that continuously counts from 0000 to 1111. Its outputs, arranged from LSB to MSB, drive a resistor ladder network (R3-R10) that performs DAC conversion. As the counter

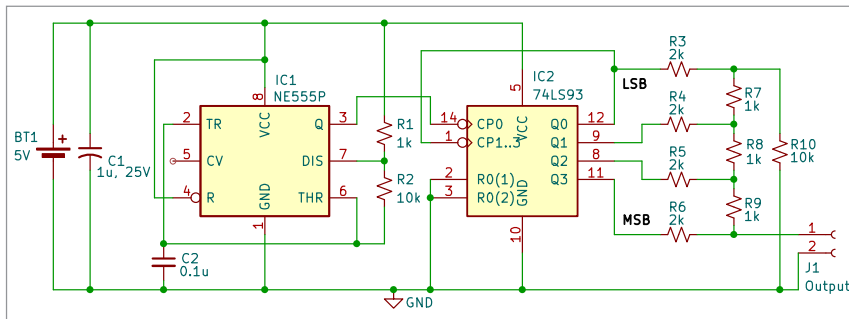


Fig. 1: Circuit diagram of the sawtooth generator

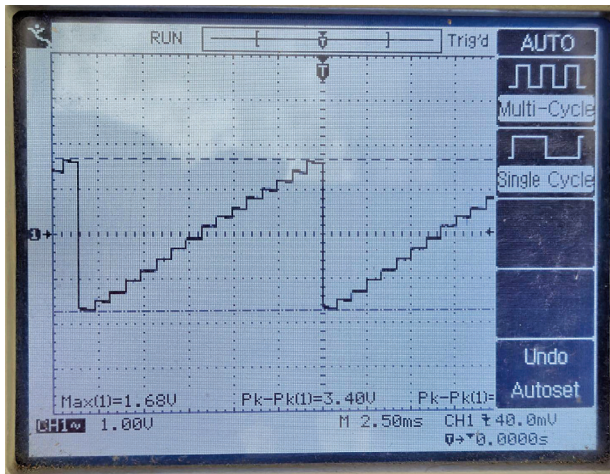


Fig. 2: Output waveform showing staircase output

result, the waveform appears stepped rather than smooth.

This system demonstrates the fundamentals of signal and pulse generation, ripple counter operation, and digital-to-analogue conversion. It also helps build practical skills in circuit assembly, debugging, and understanding noise

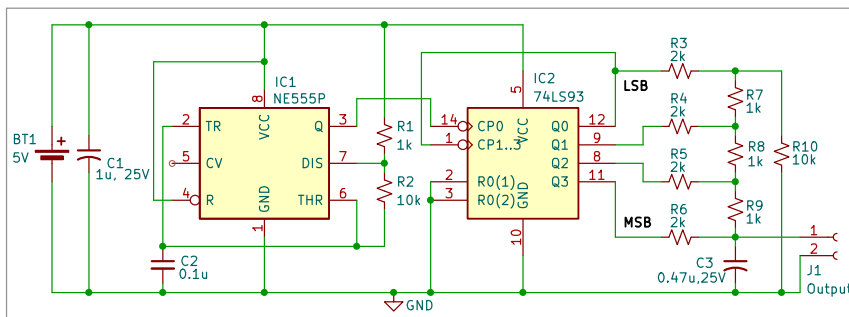


Fig. 3: Circuit diagram of the sawtooth generator after adding a capacitor across the output